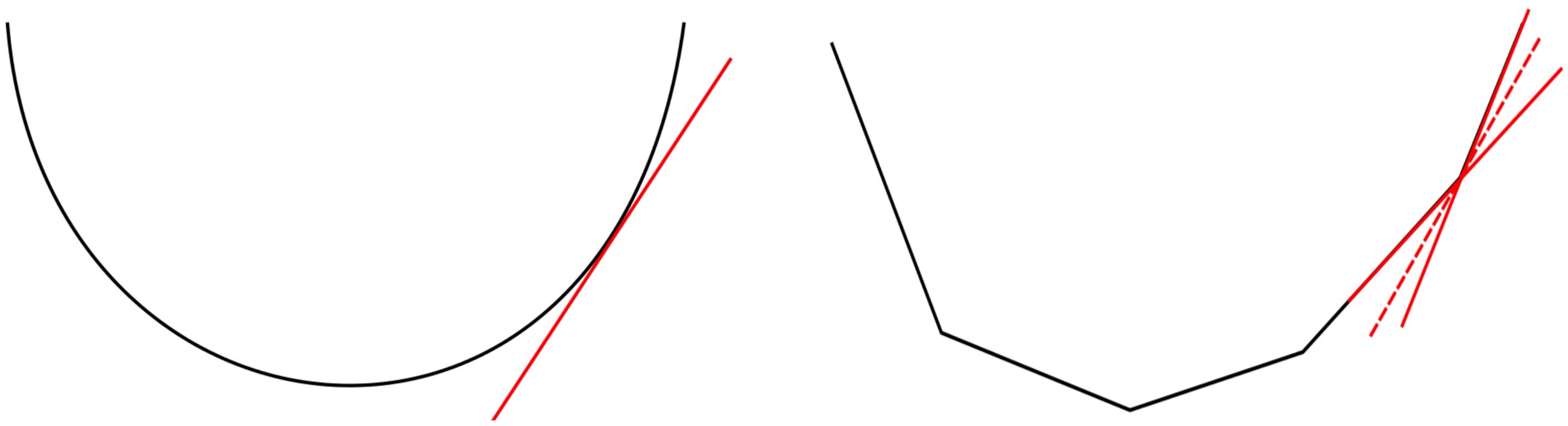
Sub-Gradient: Definition

f – convex:

$$\text{Let for all } x:\ f(x) \geq f(x^0) + \langle g, x - x^0 \rangle$$

$$g - \text{subgradient of } f \text{ in } x^0 \in \mathrm{dom} f\ .$$

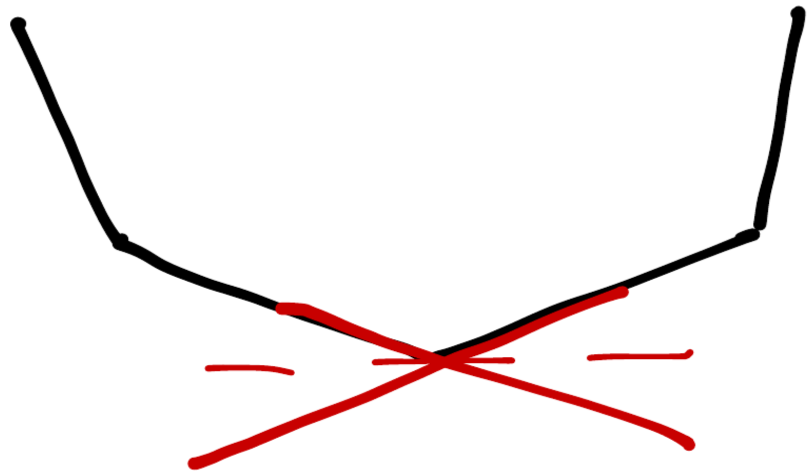


$\partial f(x) - \text{set of all subgradients in } x:$ convex, closed and bounded

f – convex, differentiable in *x*: $\partial f(x) = \{\nabla f(x)\}$

Optimality For Non-Smooth Convex Functions

Let *f* be convex, *x** - optimum of *f* iff $0 \in \partial f(x^*)$



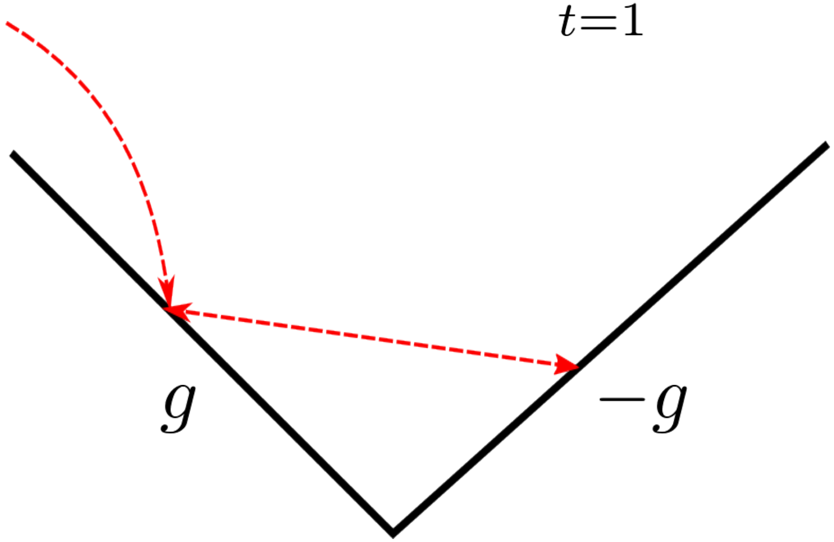
Non-zero sub-gradients can exist even in the optimum!

Sub-Gradient Method

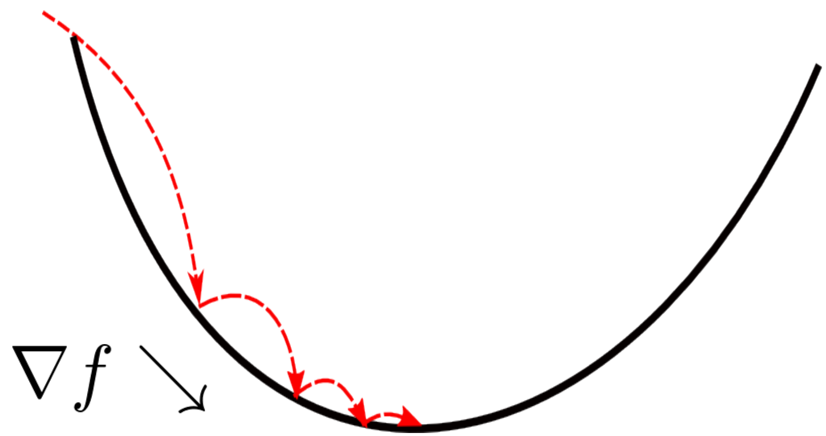
Let *f* be convex, Lipschitz continuous and $\|x^0 - x^*\| \leq R$

$$x^{t+1} = x^t - \alpha^t g(x^t)$$

$$\alpha_t > 0, \;\; \alpha^t \xrightarrow{t \to \infty} 0, \;\; \sum_{t=1}^{\infty} \alpha_t = \infty \qquad \text{e.g. } \alpha^t \propto \frac{1}{t}, \frac{1}{\sqrt{t}}$$



Constant step-size



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Sub-Gradient Method

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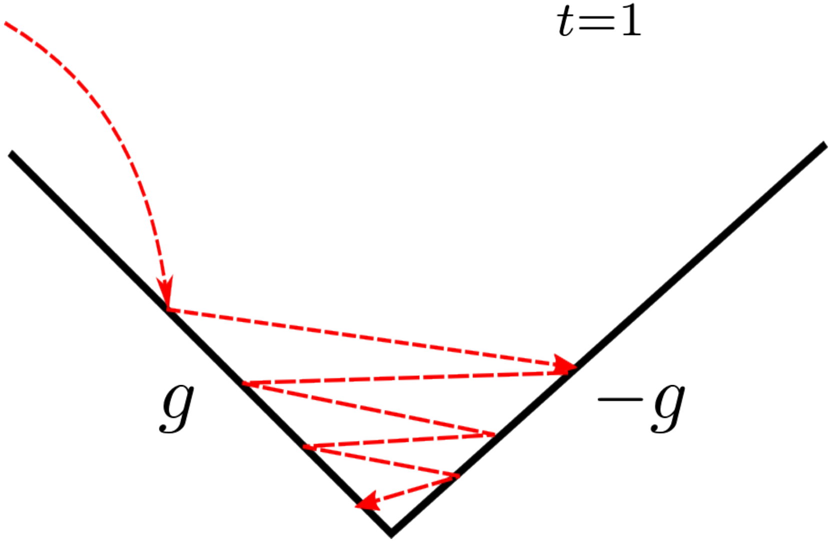
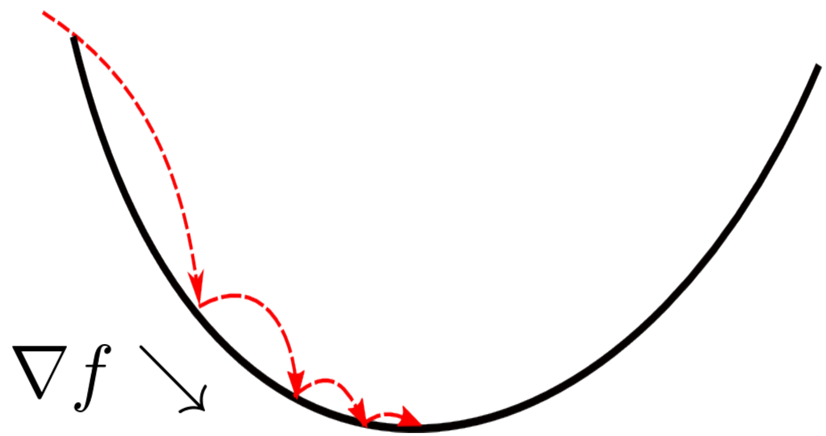


Diminishing step-size



Constant step-size

Sub-Gradient Method

Let *f* be convex, Lipschitz continuous and $\|x^0 - x^*\| \leq R$

$$x^{t+1} = x^t - \alpha^t g(x^t) \qquad \alpha_t > 0,\ \alpha^t \rightarrow \ldots$$

Improvement of the distance to the optimum

$$\|x^{t+1} - x^*\| \leq \|x^t - x^*\|$$

for: $0 < \alpha^t < 2\dfrac{f(x^t) - f(x^*)}{\|g^t\|}$

Sub-Gradient Method

Let *f* be convex, Lipschitz continuous and $\|x^0 - x^*\| \leq R$

$$x^{t+1} = x^t - \alpha^t g(x^t) \qquad \alpha_t > 0,\ \alpha^t \to \dots$$

Improvement of the distance to the optimum

$$\|x^{t+1} - x^*\| \leq \|x^t - x^*\|$$

for: $0 < \alpha^t < 2\dfrac{f(x^t) - f(x^*)}{\|g^t\|}$

Total improvement of the objective:

$$\epsilon := f^t - f^* \leq \frac{MR}{\sqrt{t+1}}$$

(for $\alpha^t = \dfrac{R}{\|g^t\|\sqrt{t+1}}$)

Sub-Gradient Method

Let *f* be convex, Lipschitz continuous and $\|x^0 - x^*\| \leq R$

$$x^{t+1} = x^t - \alpha^t g(x^t) \qquad \alpha_t > 0,\ \alpha^t \to \ldots$$

Improvement of the distance to the optimum $\quad \|x^{t+1} - x^*\| \leq \|x^t - x^*\|$

for: $\quad 0 < \alpha^t < 2\frac{f(x^t) - f(x^*)}{\|g^t\|}$

Total improvement of the objective: $\quad \epsilon := f^t - f^* \leq \frac{MR}{\sqrt{t+1}}$

(for $\quad \alpha^t = \frac{R}{\|g^t\|\sqrt{t+1}}$)

Convergence rate: $O(\frac{1}{\sqrt{t}})$ or $O(\frac{1}{\epsilon^2})$

$\epsilon = 0.1 \quad t = 100$

$\epsilon = 0.01 \quad t = 10000$

Gradient vs. Sub-Gradient Methods

Gradient descent:

$$x^{t+1} = x^t - \alpha^t \nabla f(x^t)$$

$$\alpha^t = \frac{1}{L}$$

$$\|x^{t+1} - x^*\|^2 \le \|x^t - x^*\|^2 - \frac{1}{L^2}\|\nabla f(x^t)\|^2$$

$$f(x^{t+1}) \le f(x^t) - \frac{1}{2L}\|\nabla f\|^2$$

$$f(x^t) - f^* \le \frac{2L\|x^0 - x^*\|}{t+4}$$

$$O(\frac{L}{\epsilon}) \quad \epsilon = 0.1 \quad t = 10$$

$$\epsilon = 0.01 \quad t = 100$$

Sub-gradient method:

$$x^{t+1} = x^t - \alpha^t g(x^t)$$

$$\alpha_t > 0, \ \alpha^t \xrightarrow{t \to \infty} 0, \ \sum_{t=1}^{\infty} \alpha_t = \infty$$

$$\|x^{t+1} - x^*\| \le \|x^t - x^*\|$$

$$f^t - f^* \le \frac{MR}{\sqrt{t+1}}$$

$$O(\frac{1}{\epsilon^2}) \quad \epsilon = 0.1 \quad t = 100$$

$$\epsilon = 0.01 \quad t = 10000$$

Gradient descent:

$$x^{t+1} = x^t - \alpha^t \nabla f(x^t)$$

$$\alpha^t = \frac{1}{L}$$

$$\|x^{t+1} - x^*\|^2 \leq \|x^t - x^*\|^2 - \frac{1}{L^2}\|\nabla f(x^t)\|^2$$

$$f(x^{t+1}) \leq f(x^t) - \frac{1}{2L}\|\nabla f\|^2$$

$$f(x^t) - f^* \leq \frac{2L\|x^0 - x^*\|}{t+4}$$

$$O(\frac{L}{\epsilon}) \quad \begin{array}{ll} \epsilon = 0.1 & t = 10 \\ \epsilon = 0.01 & t = 100 \end{array}$$

Sub-gradient method:

$$x^{t+1} = x^t - \alpha^t g(x^t)$$

$$\alpha_t > 0, \ \alpha^t \xrightarrow{t \to \infty} 0, \ \sum_{t=1}^{\infty} \alpha_t = \infty$$

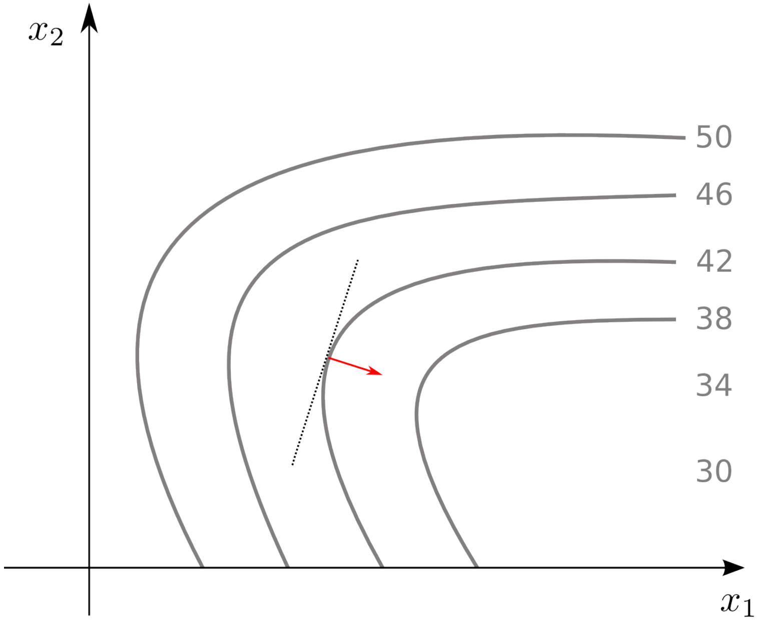
$$\|x^{t+1} - x^*\| \leq \|x^t - x^*\|$$

$$f^t - f^* \leq \frac{MR}{\sqrt{t+1}}$$

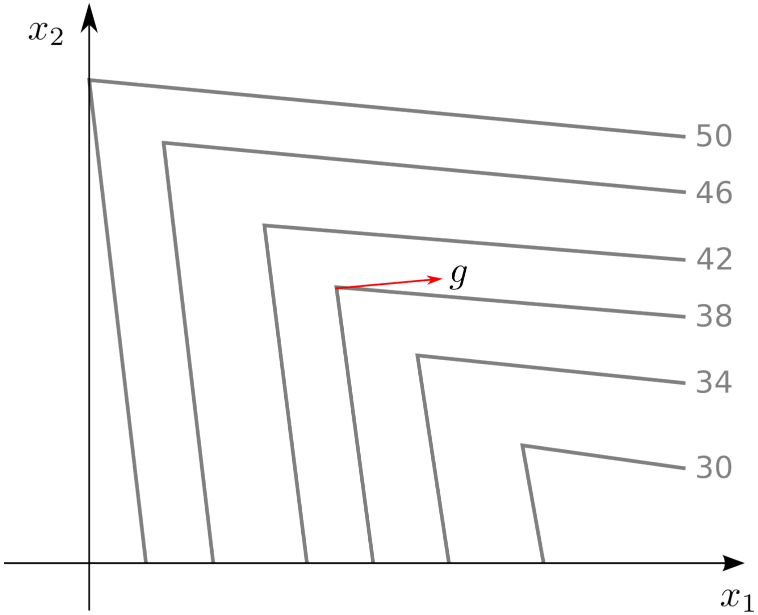
$$O(\frac{1}{\epsilon^2}) \quad \begin{array}{ll} \epsilon = 0.1 & t = 100 \\ \epsilon = 0.01 & t = 10000 \end{array}$$

Smooth function:

$$f(x^{t+1}) \leq f(x^t) - \frac{1}{2L}\|\nabla f\|^2$$



Non-smooth function:



Sub-Gradient Method

Let f be differentiable and ∇f be Lipschitz-continuous

$$x^{t+1} = x^t - \alpha^t g(x^t) \qquad \alpha_t > 0,\ \alpha^t \to \dots$$

$$x^{t+1} = x^t - \frac{1}{L} \nabla f(x^t)$$

What converges faster, gradient or sub-gradient method?

1) Gradient
2) sub-gradient
3) Depends on the sub-gradient step-size
4) Sub-gradient method can not be used for differentiable functions
5) No idea

Super-Gradient for the Dual LP

$$D(\phi) = \sum_{v \in \mathcal{V}} \min_{x_v \in \mathcal{X}_v} \Big[\overbrace{\theta_v(x_v) - \sum_{u \in \mathcal{N}_b(v)} \phi_{v,u}(x_v)}^{\theta_v^\phi(x_v)} \Big] + \sum_{uv \in \mathcal{E}} \min_{(x_u,x_v) \in \mathcal{X}_{uv}} \underbrace{[\theta_{uv}(x_u, x_v) + \phi_{v,u}(x_v) + \phi_{u,v}(x_u)]}_{\theta_{uv}^\phi(x_u,x_v)}$$

$$\text{What about } \frac{\partial D}{\partial \phi}?$$

Super-Gradient for the Dual LP

$$D(\phi) = \sum_{v\in\mathcal{V}} \min_{x_v\in\mathcal{X}_v} \Big[\overbrace{\theta_v(x_v) - \sum_{u\in\mathcal{N}_b(v)} \phi_{v,u}(x_v)}^{\theta_v^\phi(x_v)}\Big]$$

$$+ \sum_{uv\in\mathcal{E}} \min_{(x_u,x_v)\in\mathcal{X}_{uv}} \underbrace{[\theta_{uv}(x_u,x_v) + \phi_{v,u}(x_v) + \phi_{u,v}(x_u)]}_{\theta_{uv}^\phi(x_u,x_v)}$$

$$\text{What about } \frac{\partial D}{\partial \phi}?$$

$$\tilde{D}(\phi) = \min_x \tilde{\theta}(x,\phi) \quad \tilde{\theta} - \text{differentiable on } \phi$$

$$\text{Then} \quad \partial\tilde{D}(\phi) = \text{conv}\{\nabla\tilde{\theta}(x^*,\phi)\colon\, x^* \in \arg\min_x \tilde{\theta}(x,\phi)\}$$

Super-Gradient for the Dual LP

$$D(\phi) = \sum_{v \in \mathcal{V}} \overbrace{\left[\theta_v(x'_v) - \sum_{u \in \mathcal{N}_b(v)} \phi_{v,u}(x'_v)\right]}^{\theta_v^\phi(x'_v)}$$

$$+ \sum_{uv \in \mathcal{E}} \underbrace{[\theta_{uv}(x''_u, x''_v) + \phi_{v,u}(x''_v) + \phi_{u,v}(x''_u)]}_{\theta_{uv}^\phi(x''_u, x''_v)}$$

$$x' = \arg \min_{x_v \in \mathcal{X}_v} \theta_v^\phi(x_v) \qquad (x''_u, x''_v) = \arg \min_{(x_u, x_v) \in \mathcal{X}_{uv}} \theta_{uv}^\phi(x_u, x_v)$$

Super-Gradient for the Dual LP

$$D(\phi) = \sum_{v \in \mathcal{V}} \overbrace{\left[\theta_v(x'_v) - \sum_{u \in \mathcal{N}_b(v)} \phi_{v,u}(x'_v)\right]}^{\theta_v^\phi(x'_v)}$$

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$$\frac{\partial D}{\partial \phi_{v,u}(s)} = -1 \cdot [\![s = x'_v]\!] + 1 \cdot [\![s = x''_v]\!]$$

Super-Gradient for the Dual LP

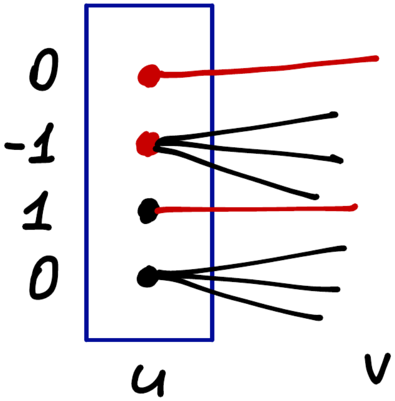
$$D(\phi) = \sum_{v\in\mathcal{V}} \overbrace{\left[\theta_v(x'_v) - \sum_{u\in\mathcal{N}_b(v)} \color{red}{\phi_{v,u}(x'_v)}\right]}^{\theta^{\phi}_v(x'_v)}$$

$$+ \sum_{uv\in\mathcal{E}} \underbrace{[\theta_{uv}(x''_u, x''_v) + \phi_{v,u}(x''_v) + \color{red}{\phi_{u,v}(x''_u)}]}_{\theta^{\phi}_{uv}(x''_u, x''_v)}$$

$$x' = \arg\min_{x_v\in\mathcal{X}_v} \theta^{\phi}_v(x_v) \qquad (x''_u, x''_v) = \arg\min_{(x_u,x_v)\in\mathcal{X}_{uv}} \theta^{\phi}_{uv}(x_u, x_v)$$

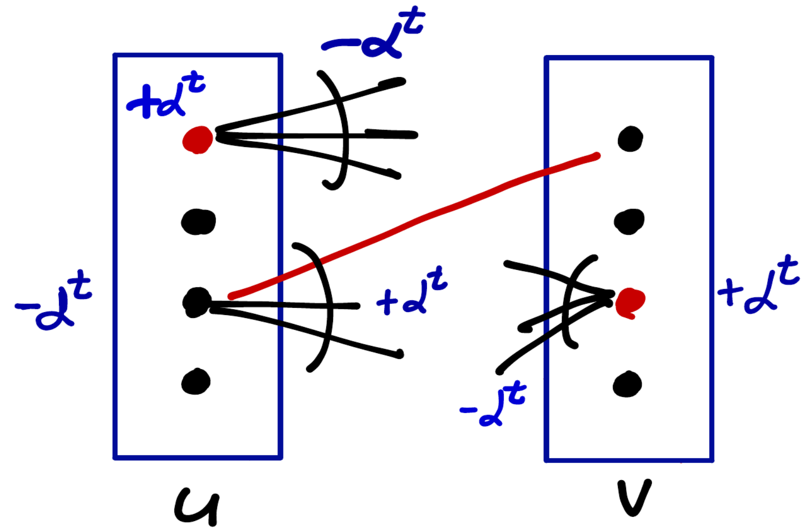
$$\frac{\partial D}{\partial \phi_{v,u}(s)} = -1\cdot [\![s = x'_v]\!] + 1\cdot [\![s = x''_v]\!]$$

$$\frac{\partial D}{\partial \phi_{v,u}(s)} = \begin{cases} 0, & s \neq x'_v \quad \text{and} \quad s \neq x''_v \\ 0, & = \quad \text{and} \quad = \\ -1, & = \quad \text{and} \quad \neq \\ 1, & \neq \quad \text{and} \quad =, \end{cases}$$



Super-Gradient Method for the Dual LP

$$\phi^{t+1} = \phi^t + \alpha^t g(\phi^t)$$



super-gradient: $x'_v = \operatorname*{argmin}_{x_v \in \mathcal{X}_v} \theta^\phi_v(x_v)$ and $(x''_u, x''_v) = \operatorname*{argmin}_{x_{uv} \in \mathcal{X}_{uv}} \theta^\phi_{uv}(x_u, x_v)$

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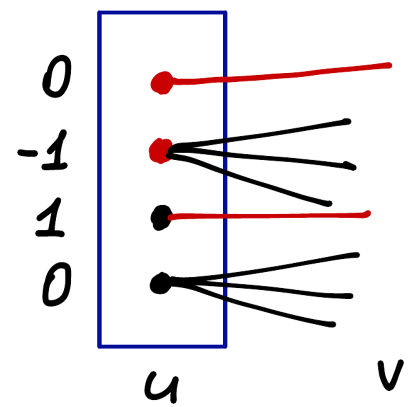
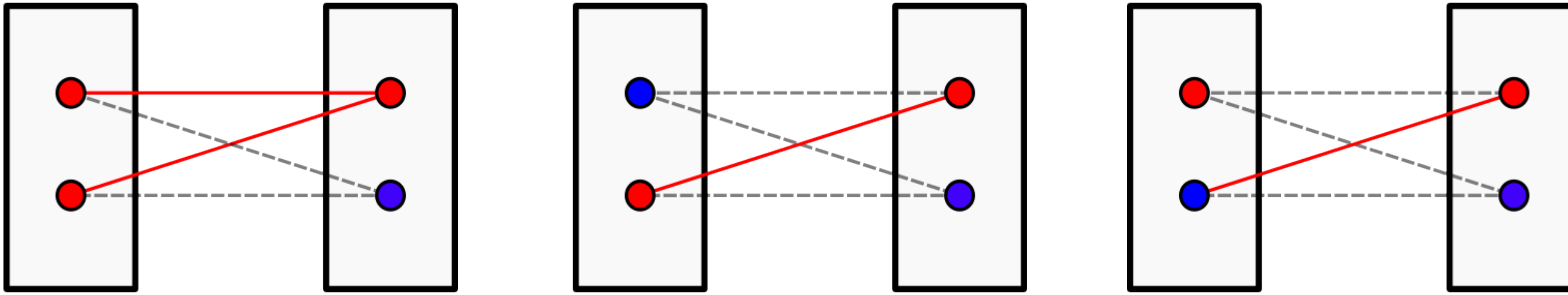
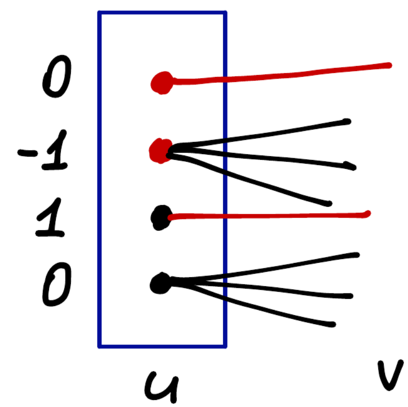




$\text{mi}[\theta]$ $\qquad \dfrac{\partial D}{\partial \phi} = 0$ $\qquad \dfrac{\partial D}{\partial \phi} \neq 0$

super-gradient: $x'_v = \underset{x_v \in \mathcal{X}_v}{\operatorname{argmin}}\ \theta^{\phi}_v(x_v)$ and $(x''_u, x''_v) = \underset{x_{uv} \in \mathcal{X}_{uv}}{\operatorname{argmin}}\ \theta^{\phi}_{uv}(x_u, x_v)$

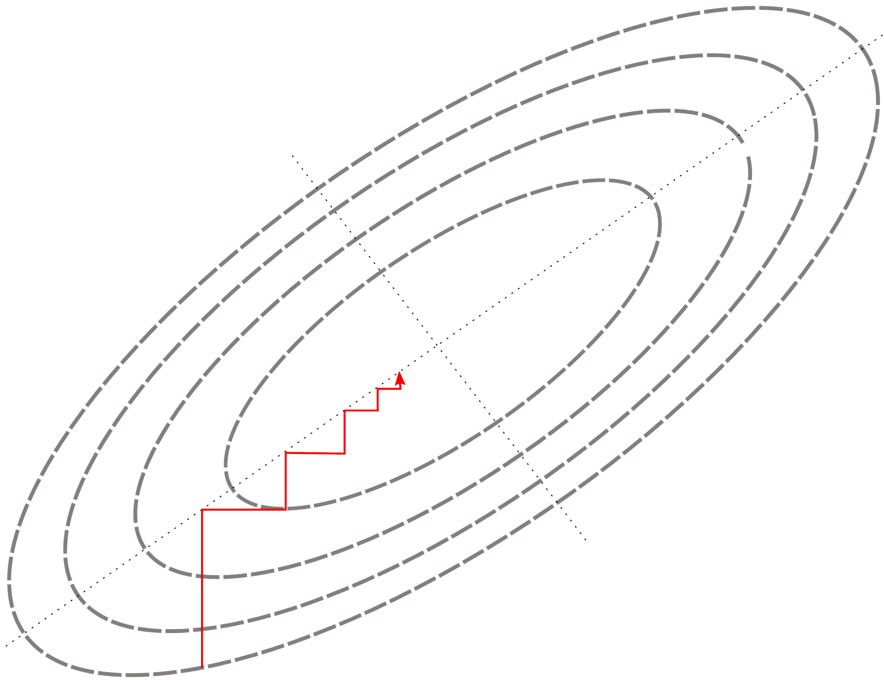
$$\frac{\partial D}{\partial \phi_{v,u}(s)} = \begin{cases} 0, & s \neq x'_v \quad \text{and} \quad s \neq x''_v \\ 0, & = \quad \text{and} \quad = \\ -1, & = \quad \text{and} \quad \neq \\ 1, & \neq \quad \text{and} \quad =, \end{cases}$$



Cyclic (Gauss-Seidel) Coordinate Descent

$$x_i^{t+1} = \arg\min_{\xi} f(x_1^{t+1}, \ldots, x_{i-1}^{t+1}, \xi, x_{i+1}^{t}, \ldots, x_n^{t})$$

$$i = i + 1 \mod n$$



Cyclic (Gauss-Seidel) Coordinate Descent

$$x_i^{t+1} = \arg\min_{\xi} f(x_1^{t+1}, \ldots, x_{i-1}^{t+1}, \xi, x_{i+1}^t, \ldots, x_n^t)$$

$$i = i + 1 \mod n$$

What does hold here:

1) $f(x^{t+1}) \leq f(x^t)$

2) $f(x^{t+1}) \geq f(x^t)$

3) None is correct

4) I do not know

Cyclic (Gauss-Seidel) Coordinate Descent

$$x_i^{t+1} = \arg\min_{\xi} f(x_1^{t+1}, \ldots, x_{i-1}^{t+1}, \xi, x_{i+1}^t, \ldots, x_n^t)$$

$$i = i + 1 \mod n$$

Let $f^* > -\infty$

What does hold here:

1) $\{f^t\}$ is converging

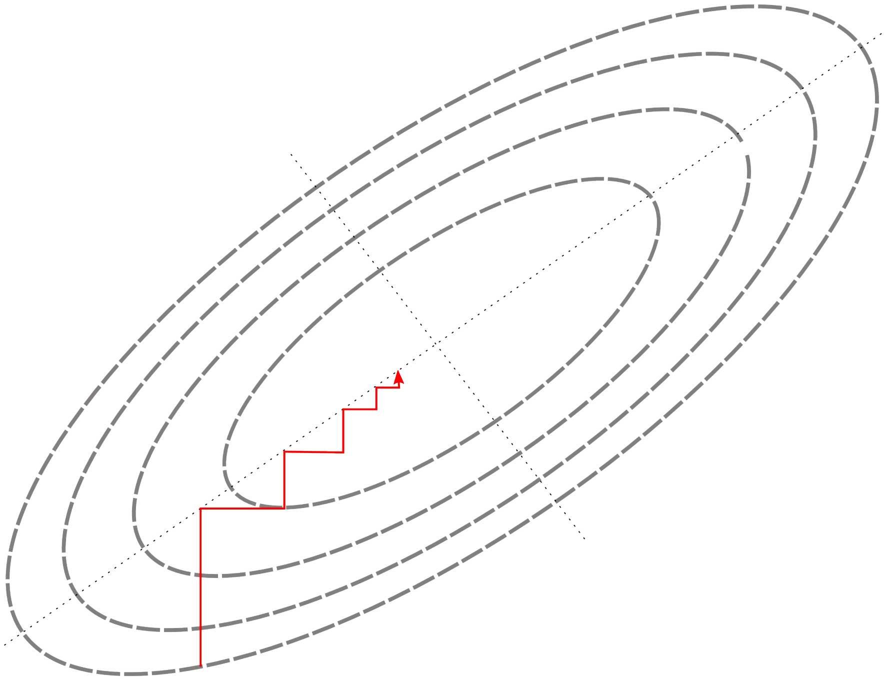
2) $\{f^t\}$ is not converging

3) None is correct

4) I do not know

Cyclic (Gauss-Seidel) Coordinate Descent

$$x_i^{t+1} = \arg\min_{\xi} f(x_1^{t+1}, \ldots, x_{i-1}^{t+1}, \xi, x_{i+1}^{t}, \ldots, x_n^{t})$$



$$i = i + 1 \mod n$$

Theorem: Let 1) *f* be continuously differentiable on $\boldsymbol{R}^n$,
2) and minimum over x_i is attained uniquely for all *i*.

Then for every limit point x^* it holds $\nabla f(x^*) = 0$

Cyclic (Gauss-Seidel) Coordinate Descent

$$x_i^{t+1} = \arg\min_{\xi} f(x_1^{t+1}, \ldots, x_{i-1}^{t+1}, \xi, x_{i+1}^t, \ldots, x_n^t)$$

$$i = i + 1 \mod n$$

Theorem: Let 1) f be continuously differentiable on $\boldsymbol{R}^n$,
2) and minimum over x_i is attained uniquely for all i.
Then for every limit point x^* it holds $\nabla f(x^*) = 0$

If $D := \{x \colon f(x) \leq f(x^0)\}$ is bounded
$\Rightarrow \{x^t\} \in D$ – bounded $\Rightarrow$ has limit points

If f – convex $\Rightarrow x^*$ – optimum.

Cyclic (Gauss-Seidel) Coordinate Descent

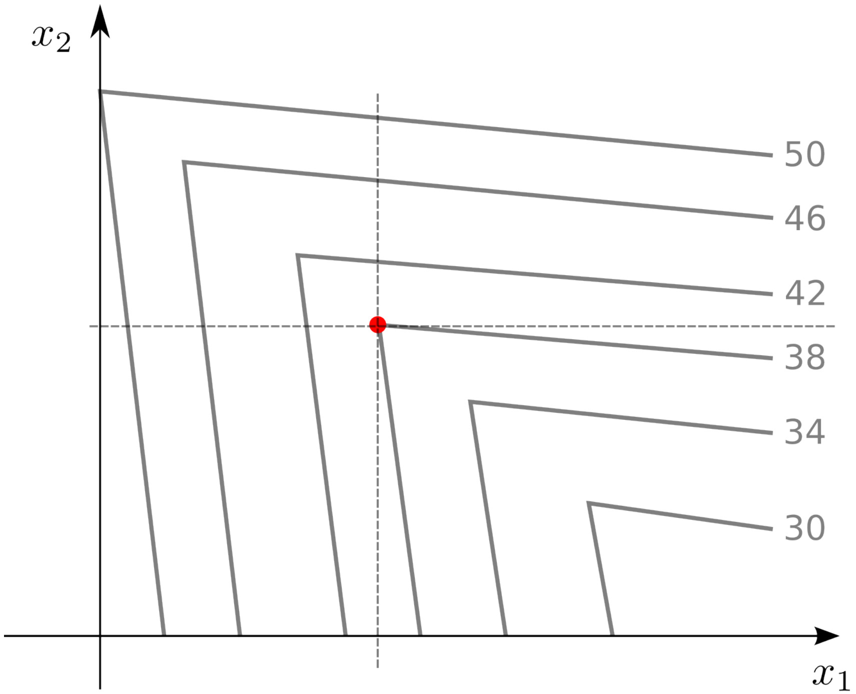
$$x_i^{t+1} = \arg\min_{\xi} f(x_1^{t+1}, \ldots, x_{i-1}^{t+1}, \xi, x_{i+1}^{t}, \ldots, x_n^{t})$$

$$i = i + 1 \mod n$$

Theorem: Let 1) f be convex, continuously differentiable on $\boldsymbol{R}^n$,
2) and minimum over x_i is attained uniquely for all i,
3) $D := \{x\colon f(x) \leq f(x^0)\}$ is bounded

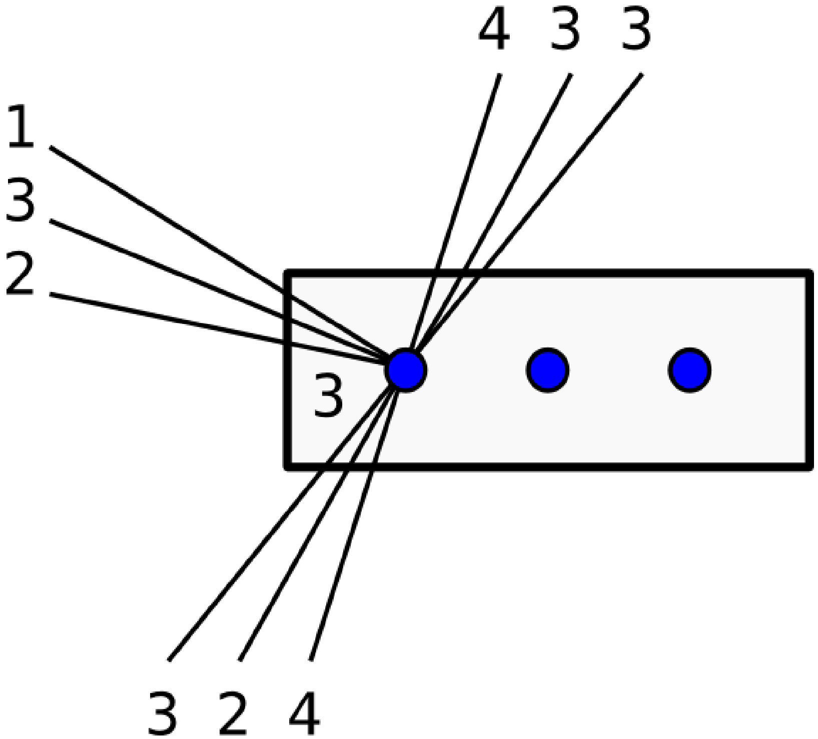
Then the algorithm converges to the optimum.

Cyclic (Gauss-Seidel) Coordinate Descent



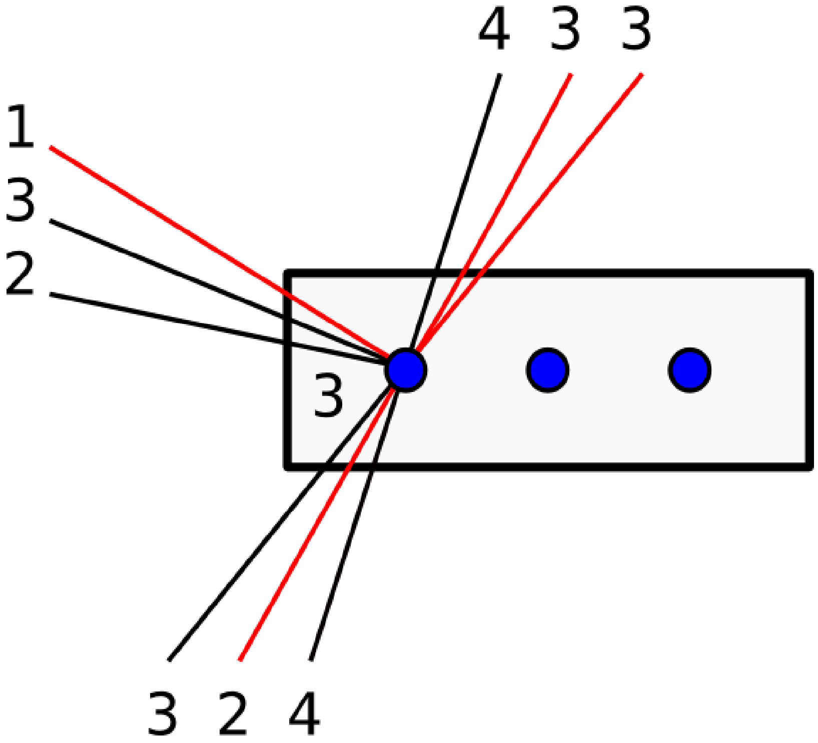
What if the function is non-smooth?

Min-Sum Diffusion Algorithm



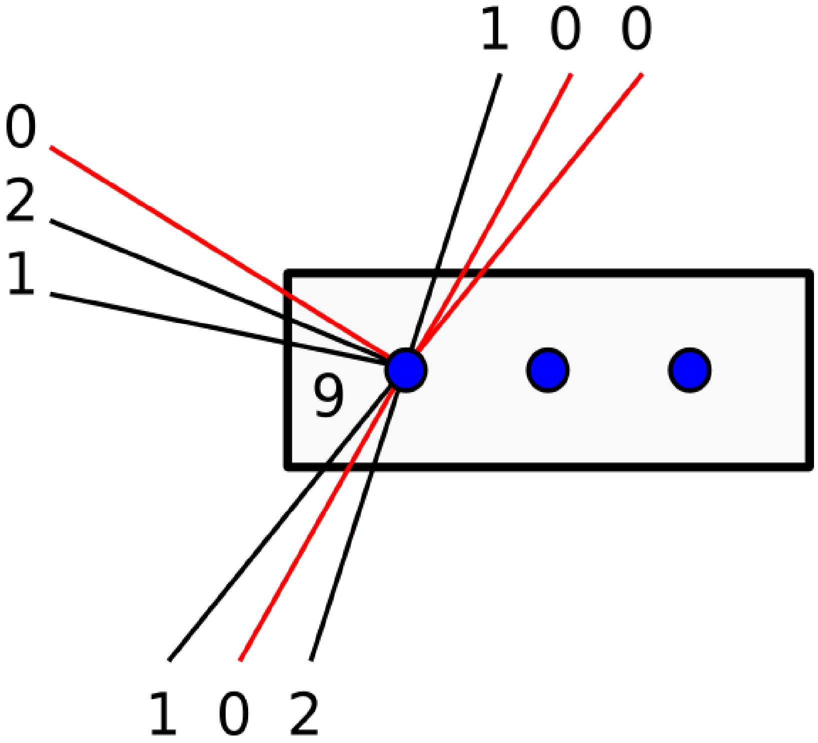
Consider the first label

Min-Sum Diffusion Algorithm



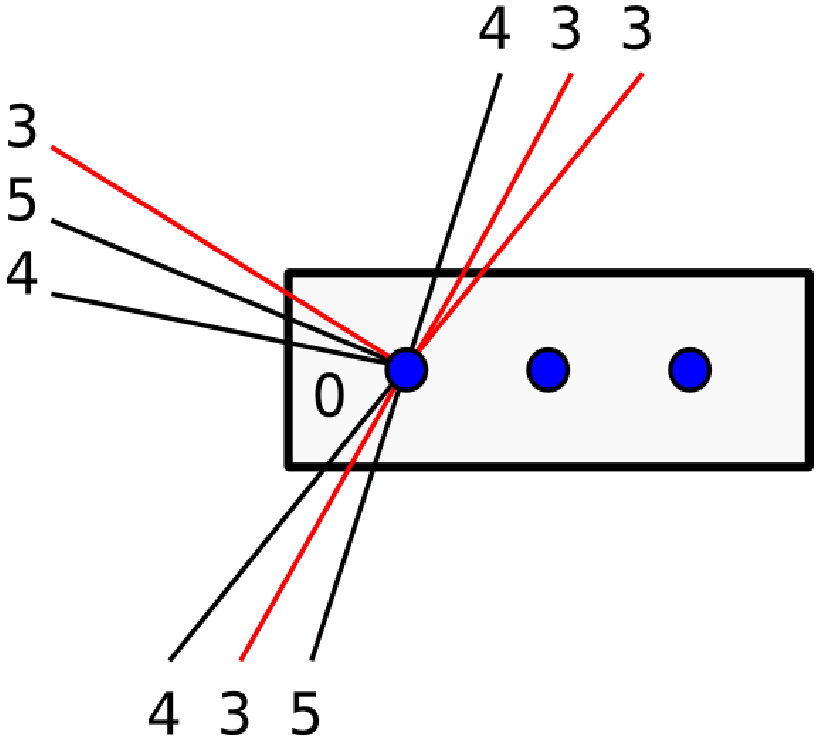
Find minimal edges in each direction

Min-Sum Diffusion Algorithm



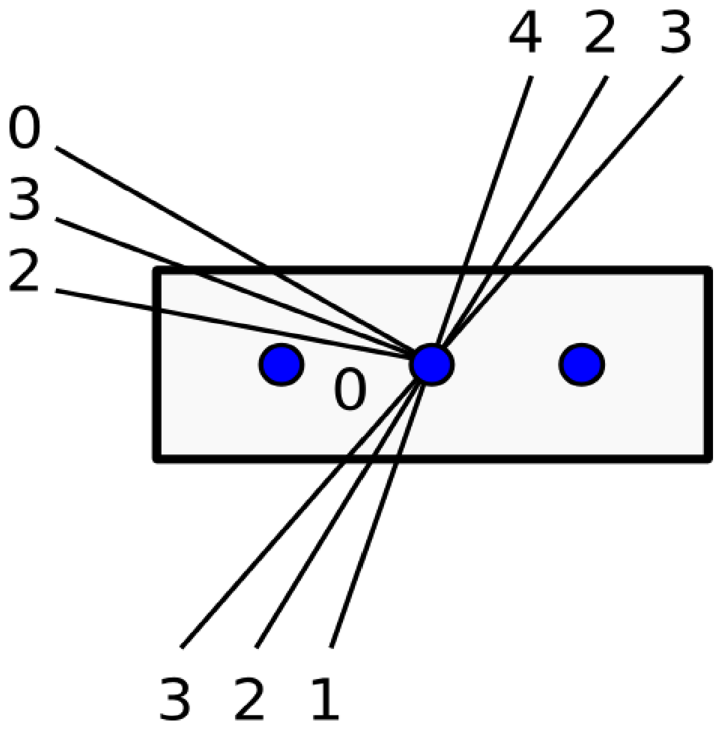
Move costs from edges to the label $\theta_u^{\phi}(x_u) += \sum_{v \in \mathcal{N}_b(v)} \min_{x_v \in \mathcal{X}_v} \theta^{\phi}(x_u, x_v)$

Min-Sum Diffusion Algorithm



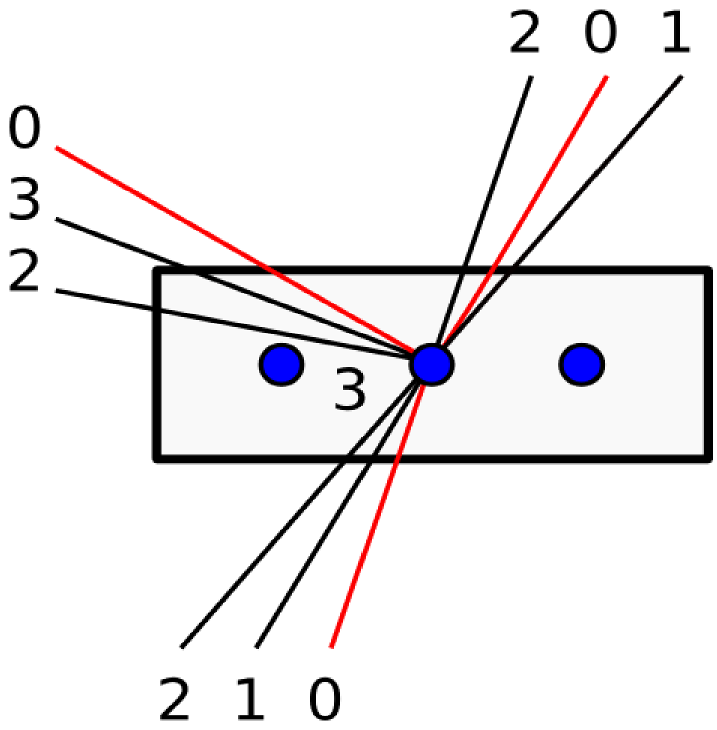
Move costs back uniformly to the edges $\theta^{\phi}_{uv}(x_u, x_v) += \theta^{\phi}_{u}(x_u)/|\mathcal{N}_b(v)|$

Min-Sum Diffusion Algorithm



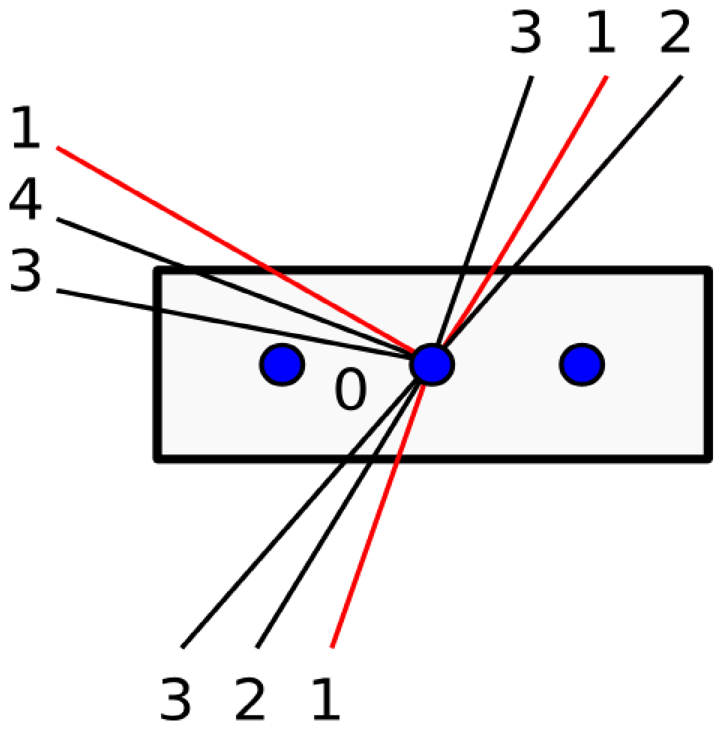
Consider the second label

Min-Sum Diffusion Algorithm



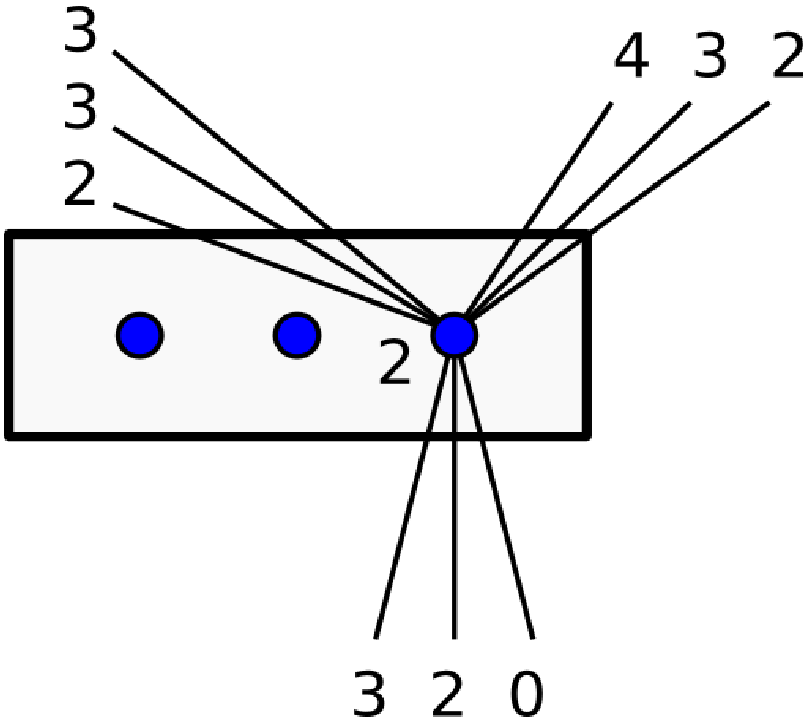
Move costs from edges to the label $\theta_u^{\phi}(x_u) {+}{=} \sum_{v \in \mathcal{N}_b(v)} \min_{x_v \in \mathcal{X}_v} \theta^{\phi}(x_u, x_v)$

Min-Sum Diffusion Algorithm



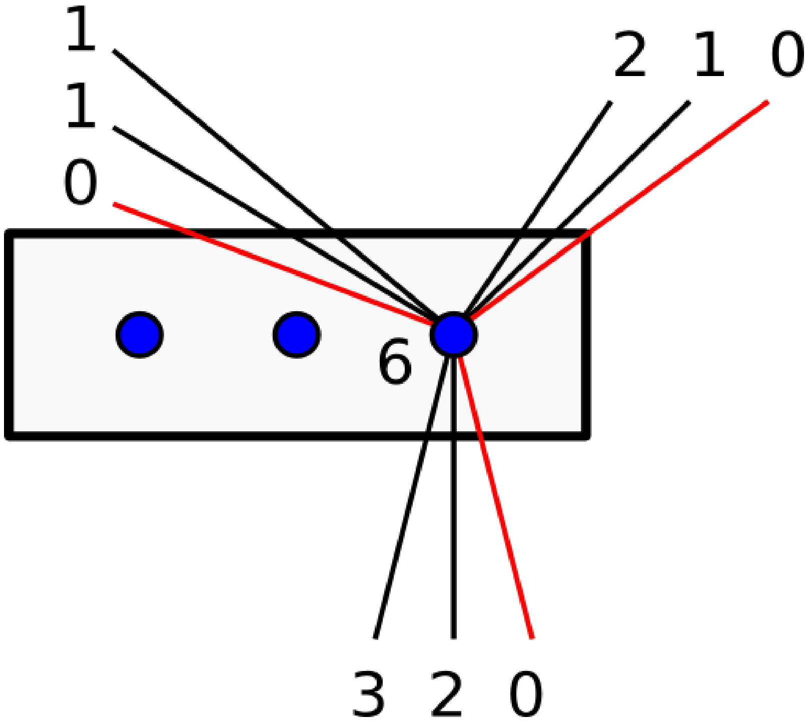
Move costs back uniformly to the edges $\theta^{\phi}_{uv}(x_u, x_v) += \theta^{\phi}_{u}(x_u)/|\mathcal{N}_b(v)|$

Min-Sum Diffusion Algorithm



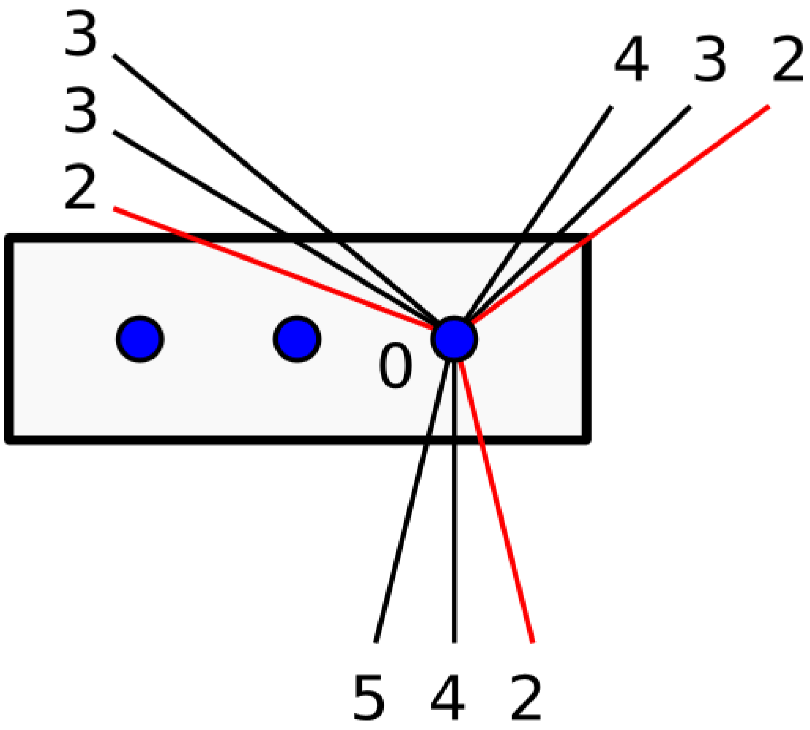
Consider the third label

Min-Sum Diffusion Algorithm



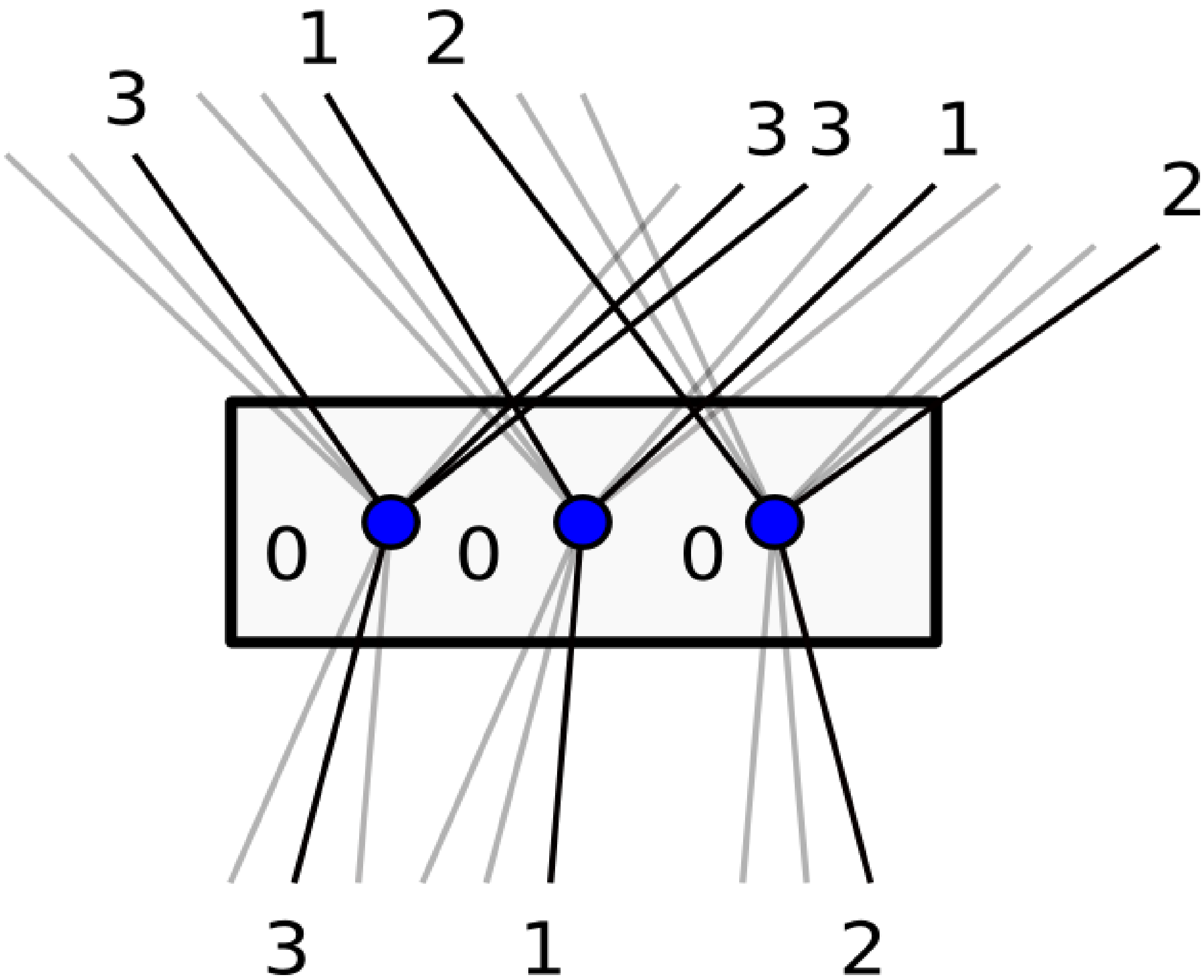
Move costs from edges to the label $\theta_u^{\phi}(x_u) += \sum_{v \in \mathcal{N}_b(v)} \min_{x_v \in \mathcal{X}_v} \theta^{\phi}(x_u, x_v)$

Min-Sum Diffusion Algorithm



Move costs back uniformly to the edges $\theta^{\phi}_{uv}(x_u, x_v) += \theta^{\phi}_{u}(x_u)/|\mathcal{N}_b(v)|$

Min-Sum Diffusion Algorithm



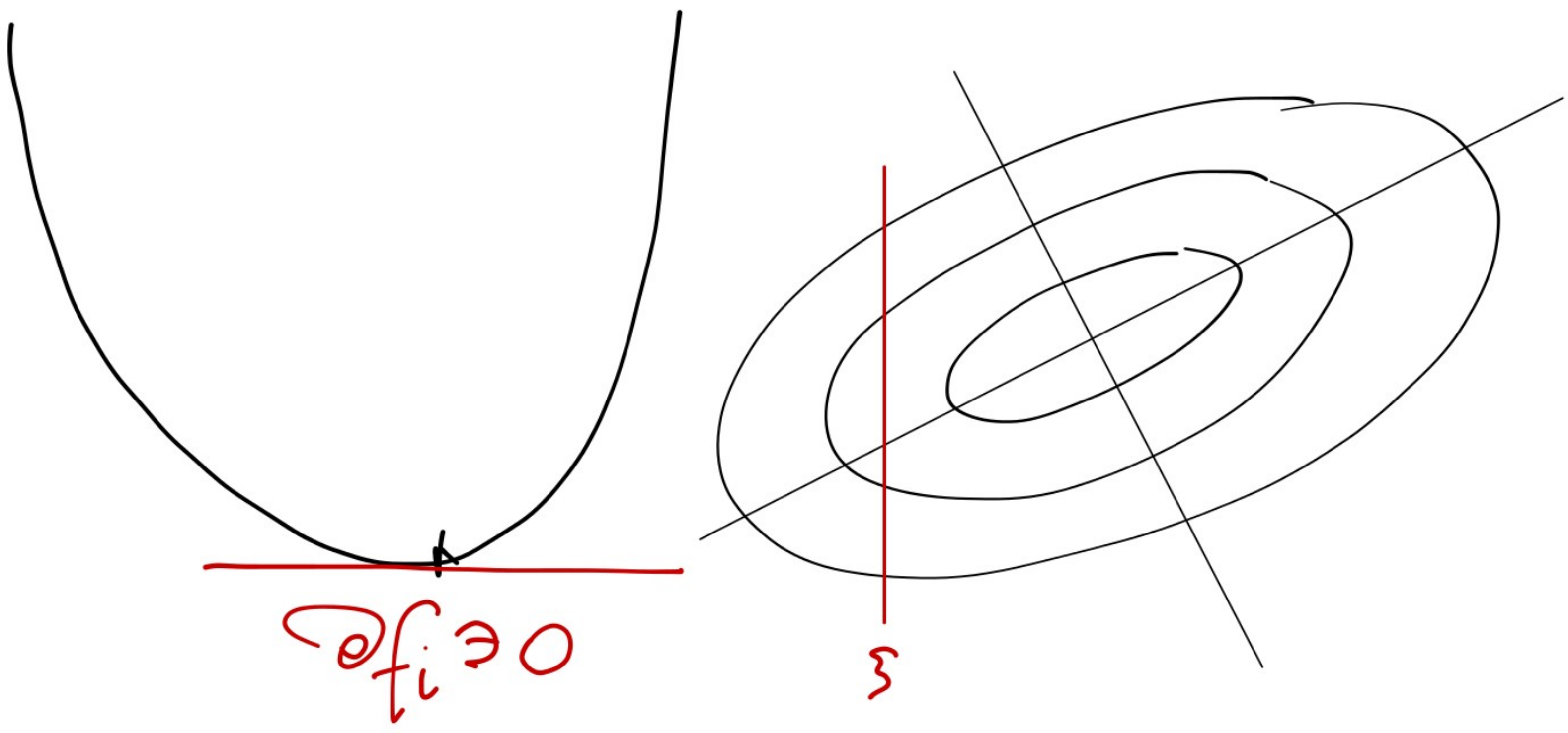
Result of the diffusion

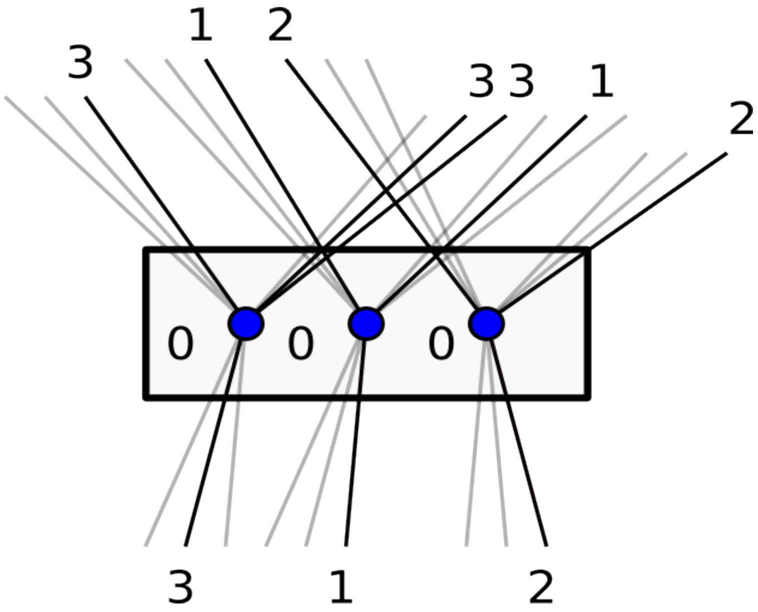
Cyclic (Gauss-Seidel) Coordinate Descent

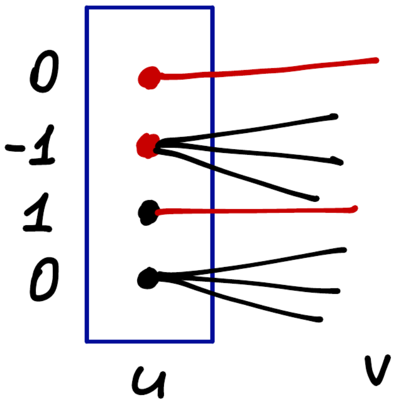
Lemma:

Let f be convex and $f_i(\xi) := f(x_1^{t+1}, \ldots, x_{i-1}^{t+1}, \xi, x_{i+1}^t, \ldots, x_n^t)$

Then f_i is convex and $\xi^* = \arg\min_\xi f_i(\xi)$ iff $\overline{0} \in \partial f_i(\xi)$.



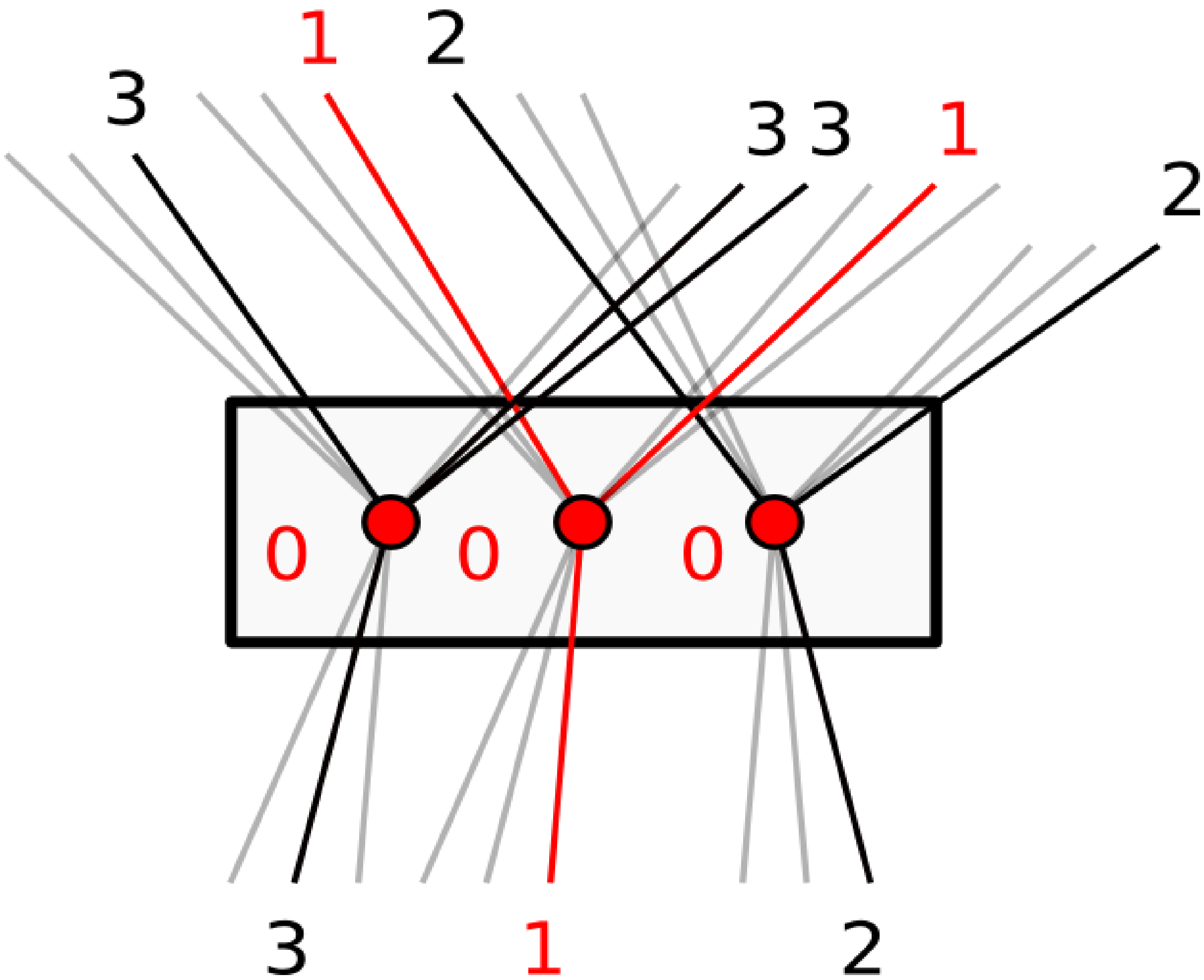




Is there a zero super-gradient?

1) Yes, corresponds to label 1
2) Yes, corresponds to label 2
3) Yes, corresponds to label 3
4) No
5) No clue

Min-Sum Diffusion Algorithm



Compute the super-gradient

Min-Sum Diffusion Algorithm: Convergence Guarantees

$$\mathrm{mi}_\epsilon[\theta]_w(x_w) = \begin{cases} 1, & \theta_w(x_w) \leq \min_{x_w \in \mathcal{X}_w} \theta_w(x_w) + \epsilon \\ 0, & \text{otherwise} \end{cases}$$

Definition: Potentials are ϵ-arc-consistent if $\mathrm{mi}_\epsilon[\theta]$ is arc-consistent

Min-Sum Diffusion Algorithm: Convergence Guarantees

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Definition: Potentials are ϵ-arc-consistent if $\mathrm{mi}_\epsilon[\theta]$ is arc-consistent

$\epsilon[\theta]$ - minimal θ for which $\mathrm{mi}_\epsilon[\theta]$ is arc-consistent

Theorem: Let ϕ^t be produced by the diffusion algorithm.
Then $\epsilon[\theta^{\phi^t}] \xrightarrow{t\to\infty} 0.$